

Unmanned ground vehicle

An **unmanned ground vehicle** (UGV) also known colloquially as **armored robot** (ARB) is a vehicle that operates while in contact with the ground without an onboard human presence. UGVs can be used for many applications where it is inconvenient, dangerous, expensive, or impossible to use an onboard human operator. Typically, the vehicle has sensors to observe the environment, and autonomously controls its behavior or uses a remote human operator to control the vehicle via teleoperation.



A Gladiator tactical unmanned ground vehicle 

The UGV is the land-based counterpart to unmanned aerial vehicles, unmanned underwater vehicles and unmanned surface vehicles. Unmanned robots are used in war^[1] and by civilians.^[2]

History

In 1904, Spanish engineer Leonardo Torres Quevedo was developing a radio-based control system, which he named *Telekino*. He chose to conduct an initial test in the form of a three-wheeled land vehicle (tricycle), which had an effective range of 20 to 30 meters, the first known example of an unmanned ground vehicle.^{[3][4]}

The first prototypes of explosive robotic drones were Aubriot-Gabet 'land torpedoes' invented in France in 1915^[5] and the Crocodile Schneider-Creusot. Twenty examples were put into service with the 2nd French Army in July 1915.^[6]

A working remote-controlled car was reported in the October 1921 issue of RCA's *World Wide Wireless* magazine. The car was controlled wirelessly via radio; it was thought the technology could be adapted to tanks.^[7] In the 1930s, the USSR developed the Teletank, a small tank, armed with a machine gun. It was remotely controlled by radio from another tank. Teletanks operated in the Winter War (1939–1940) between Finland and the USSR and at the start of the German-Soviet War after Germany invaded the USSR in 1941. During World War II, the British developed a radio-controlled version of their Matilda II infantry tank in 1941. Known as "Black Prince", it would have been used for drawing fire from concealed anti-tank guns, or for demolition missions. Due to the costs of converting the transmission system of the tank to Wilson-type gearboxes, an order for 60 tanks was cancelled.^[8]



Uran-9 unmanned ground vehicle 



RCA radio controlled car. Dayton, Ohio, 1921 

From 1942, the German Wehrmacht used the Goliath tracked mine for remote-controlled demolition work. The Goliath was a tracked vehicle carrying 60 kg of explosive charge, directed through a control cable. It modeled a miniature French tracked vehicle found after the German defeat of France in 1940. The combination of cost, low speed, reliance on a cable for control, and poor protection against weapons, meant that the Goliath was not considered a success.

The first major mobile robot development effort, named "Shakey", took place during the 1960s as a research study for the U.S. Defense Advanced Research Projects Agency (DARPA). Shakey was a wheeled platform that had a TV camera, sensors, and a computer to help guide its tasks of picking up wooden blocks and placing them in certain areas based on commands. DARPA subsequently developed a series of autonomous and semi-autonomous ground robots, often in conjunction with the U.S. Army. As part of the Strategic Computing Initiative of 1983–1993, DARPA c. 1985 demonstrated the Autonomous Land Vehicle,^[9] (ALV), the first UGV that could navigate completely autonomously on and off roads at useful speeds.^[10]

UGVs rose in significance after the full-scale Russian invasion of Ukraine and the dramatic rise in use of unmanned aerial vehicles. On March 29, 2024, as part of the Eastern Ukraine Campaign, a platoon of Russian UGVs equipped with AGS-17 automatic grenade launchers was deployed for an assault near the town of Berdychi in Ukraine, marking the first ever use of UGVs for direct frontline assaults.^[11]

During an undated operation, the 3rd Assault Brigade claimed the world's first surrender of enemy soldiers to unmanned drones in the Kharkiv Oblast. FPV and ground drones were used to attack the Russian positions. As more ground drones approached the position, the Russians opted to surrender. The Ukrainian ground drones then guided the Russians soldiers to the Ukrainian position where they surrendered. The press release read: "For the first time in history: Russian soldiers surrendered to the 3rd Assault Brigade's ground drones,".^[12]

On 22 December 2025, the 3rd Assault Brigade said that a Ukrainian UGV, a Droid TW 12.7 armed with a .50-caliber machine gun, held off Russian attacks for 45 days while being operated either remotely or using AI. The first time a position was held solely by an UGV without direct human support such as infantry.^[13]

Design

UGVs generally include: a vehicle platform, sensors, control systems, guidance interface, communication links, and systems integration features.^[14]

Platform

The platform can be a car, truck, all-terrain vehicle, etc., and includes the locomotive apparatus, sensors, and power source. Tracks, wheels, and legs are common forms of locomotion. The platform may include an articulated body and may join with other units.^{[14][15]} Power sources can be internal combustion engines, batteries, or hydrogen.^[16]